Cloud **Terraformation**

FROM CLICKOPS TO INFRASTRUCTURE AS CODE

Agenda

STORYTIME

THE LANGUAGE

THE PROVIDER

THE ASSIGNMENT

QUESTIONS

Hello from Grand Rapids!

I'm Victor Frye

Your friendly neighborhood developer

Unchallenged #1 Clippy fan

Love internet culture and content
- victorfrye.com/blog
- microsoftgraveyard.com
- shrugman.com

Passion for .NET, React, Azure, Xbox

DevOps specialist at Leading EDJE

Building with Microsoft solutions

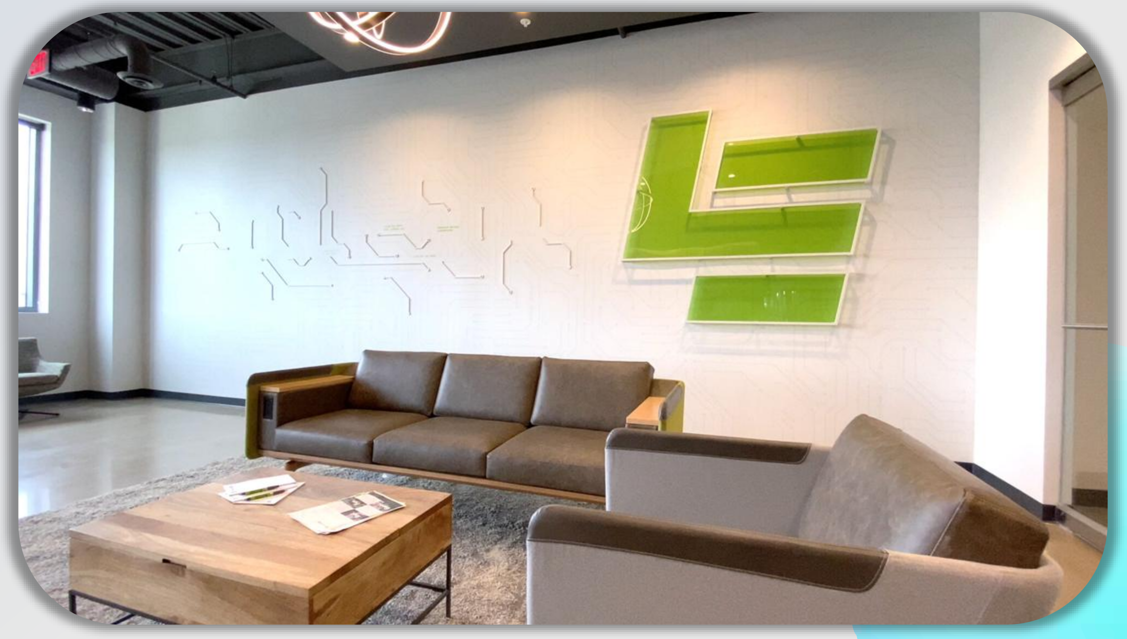
About **Leading EDJE**

Technology consultancy and services
- Strategists, innovators, partners
- Craft bespoke solutions
- Positive disruption of business norm

Real. Fun. Geeks.

FREE lunch and learns

Expert talent
- Cloud infrastructure
- Digital and app innovation
- Data and AI
- Project management



The **ClickOps story**

Build application
What infrastructure?
Click, click, deploy
Cross fingers it works
Please do not change it

The Answer

CODIFY INFRASTRUCTURE

The Language

INFRASTRUCTURE AS CODE

HashiCorp Terraform

An infrastructure as code tool

Source-available

Cloud-agnostic

Declarative

Stateful

Uses Terraform configuration language (HCL)

Elements include:
- Blocks (e.g., resources, variables, data)
- Arguments (e.g., name = “value”)
- Expressions (e.g., name = var.example)

```
resource "azurerm_resource_group" “this" {
  name     = “rg-example"
  location = "Central US"
}

resource "azurerm_static_web_app" “this" {
  name                = “stapp-example"
  resource_group_name = azurerm_resource_group.example.name
  location            = azurerm_resource_group.example.location
}
```

EXAMPLE CONFIGURATION FOR AN AZURE STATIC WEB APP

Write, Plan, Apply

WRITE	PLAN	APPLY
Define infrastructure in configuration files	Review expected changes to infrastructure	Provision infrastructure and update state

Alternative options

- Bicep
- az CLI
- CloudFormation
- Pulumi
- Ansible
- PowerShell

The Provider

YOUR CLOUD SPECIFIC INTERFACE

Multiple providers

AWS	AzureRM	Google	Kubernetes
AzAPI	AzureAD	MSGraph	GitHub

Choose your provider

AzureRM

- Automatic API versioning
- Simplicity
- Comprehensive documentation

AzAPI

- Immediate API access
- Targeted resource updates
- Fine-grained control

Terraforming Azure

- ClickOps resources in the Azure portal
- Configure workload identity federation with Entra ID
- Set up Terraform backend with blob storage
- Use template exports for Terraform
- Import existing resources into state
- Automate CI/CD with GitHub Actions

The Assignment

READY, SET, GO!

Install Terraform

DEMONSTRATION

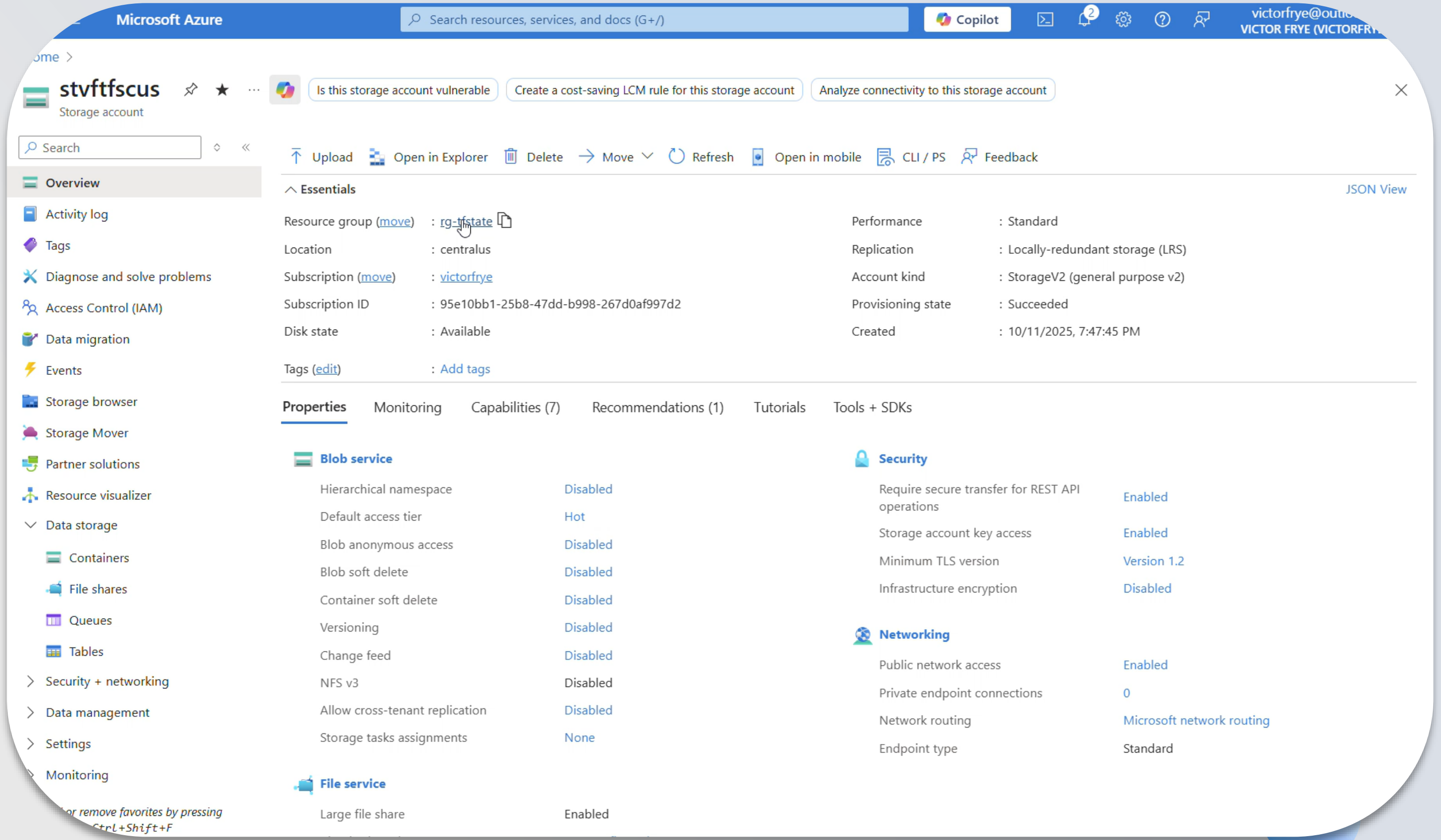
Set up identity

DEMONSTRATION

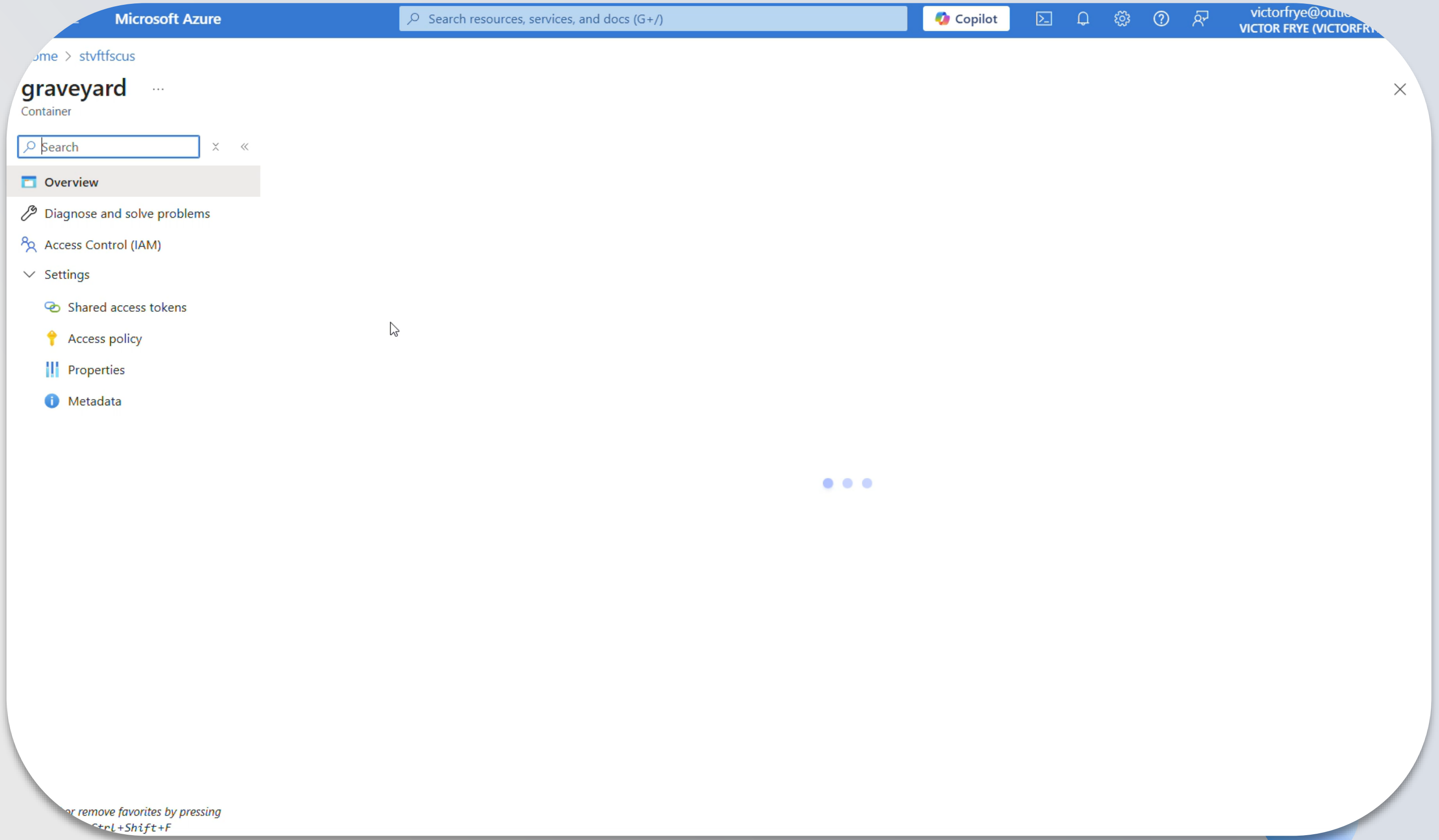
Create backend

DEMONSTRATION

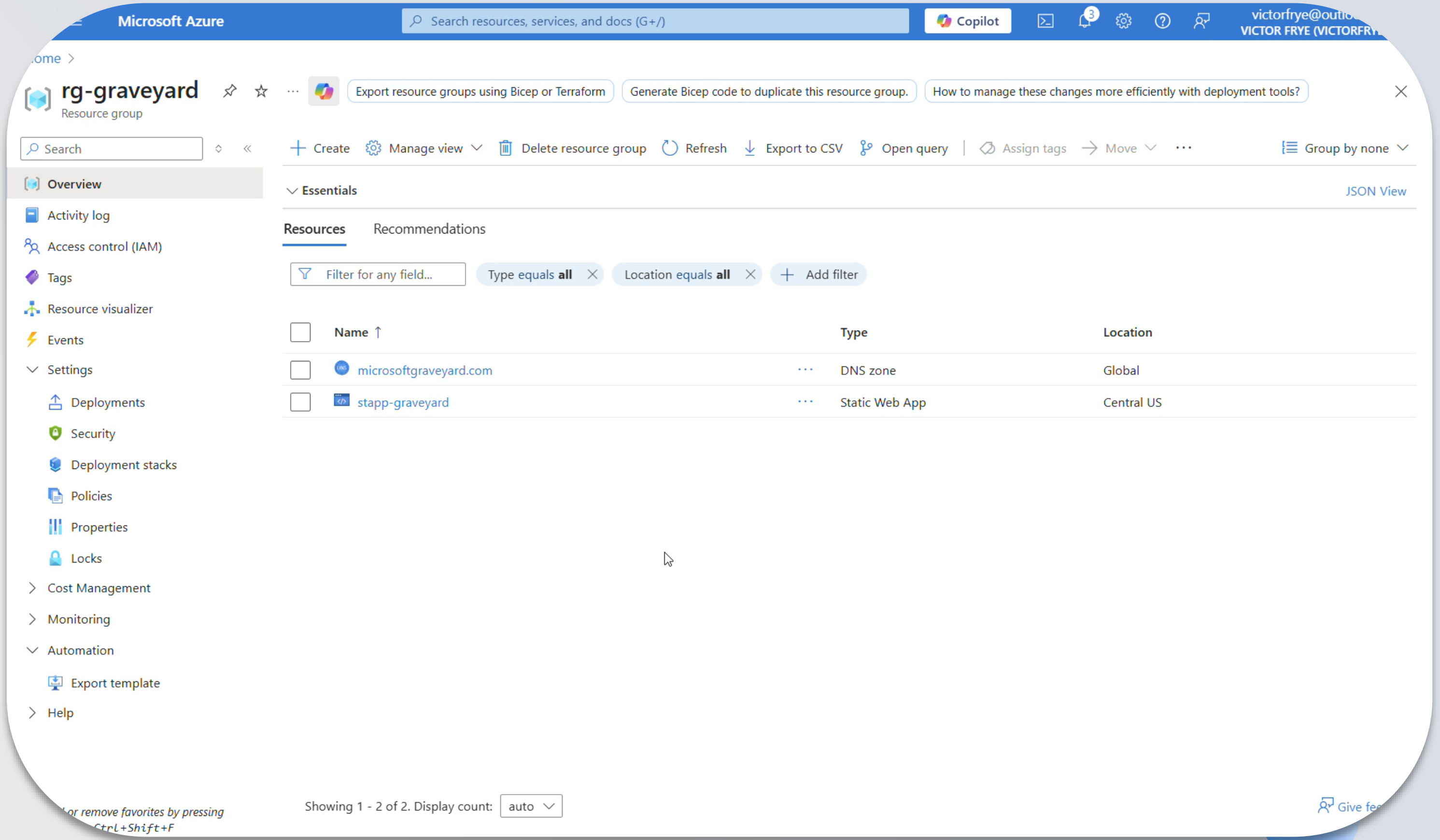
Create backend



Configure RBAC



Export Terraform



Refactor configurations

DEMONSTRATION

Import state

DEMONSTRATION

Build plan

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Apply plan

DEMONSTRATION

Automate CI/CD

DEMONSTRATION

Did it work?

SHRUGMAN.COM

Thank you

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